Do Now · retrieval

Network security, proxies and firewalls

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Can network packets travelling between the same two endpoints take different routes?

Q2

What is the transport-layer role of TCP?

Q3

What does a rule ‘allow if staff AND authenticated’ mean if authenticated is false?

Q4

Why can an open port be useful but also increase risk?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Check a firewall rule — A firewall applies configured rules to traffic according to properties such as direction, source/destination, ports and state. A basic rule example can allow TCP port 443 but deny 22.

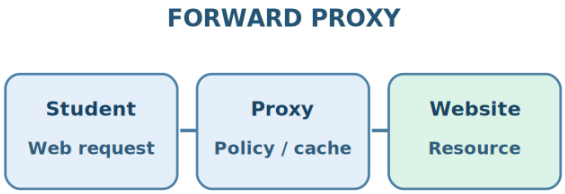
Core idea 2

Different request, different decision — A TCP connection to port 22 is denied by the illustrated policy. A firewall rule is not automatically a guarantee that the permitted service is safe.

Core idea 3

A proxy is an intermediary — A forward proxy makes requests on behalf of clients, potentially enforcing policy or caching suitable resources. HTTPS content remains encrypted unless a controlled interception mechanism is in place.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

firewall — Security control that permits or blocks network traffic according to rules.

forward proxy — Intermediary that makes requests on a client's behalf.

filter rule — A condition and action defining whether traffic is permitted.

cache — Stored copy of suitable data for reuse.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

For an inbound rule set allowing TCP 443 and denying TCP 22, decide whether two connection requests are permitted.

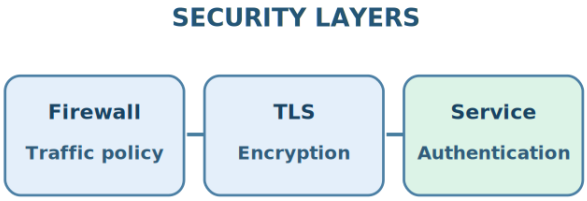
Follow the worked method

1. 443 — An inbound TCP connection destined for port 443 matches the allow rule and is permitted.

2. 22 — An inbound TCP connection destined for port 22 matches the deny rule and is blocked.

3. Scope — An allow rule permits traffic; it does not prove that an application or connection is harmless.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Allowed resource
The student's request reaches the proxy, which permits it and contacts the website; a suitable cached response may be used instead.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

A firewall allows inbound TCP 443 but denies inbound TCP 22. Is an inbound connection to destination port 22 allowed? Explain. [2]

Question 2

Name and explain two different functions of a forward proxy in a school network. [4]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Why is the claim ‘a firewall makes all allowed traffic safe’ incorrect? Explain two risks that can remain on allowed connections and identify two additional safeguards. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A forward proxy and firewall are identical because both can restrict access.

Use key vocabulary from memory

1. Explain firewall in your own words.

2. Explain forward proxy in your own words.

3. Explain filter rule in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.